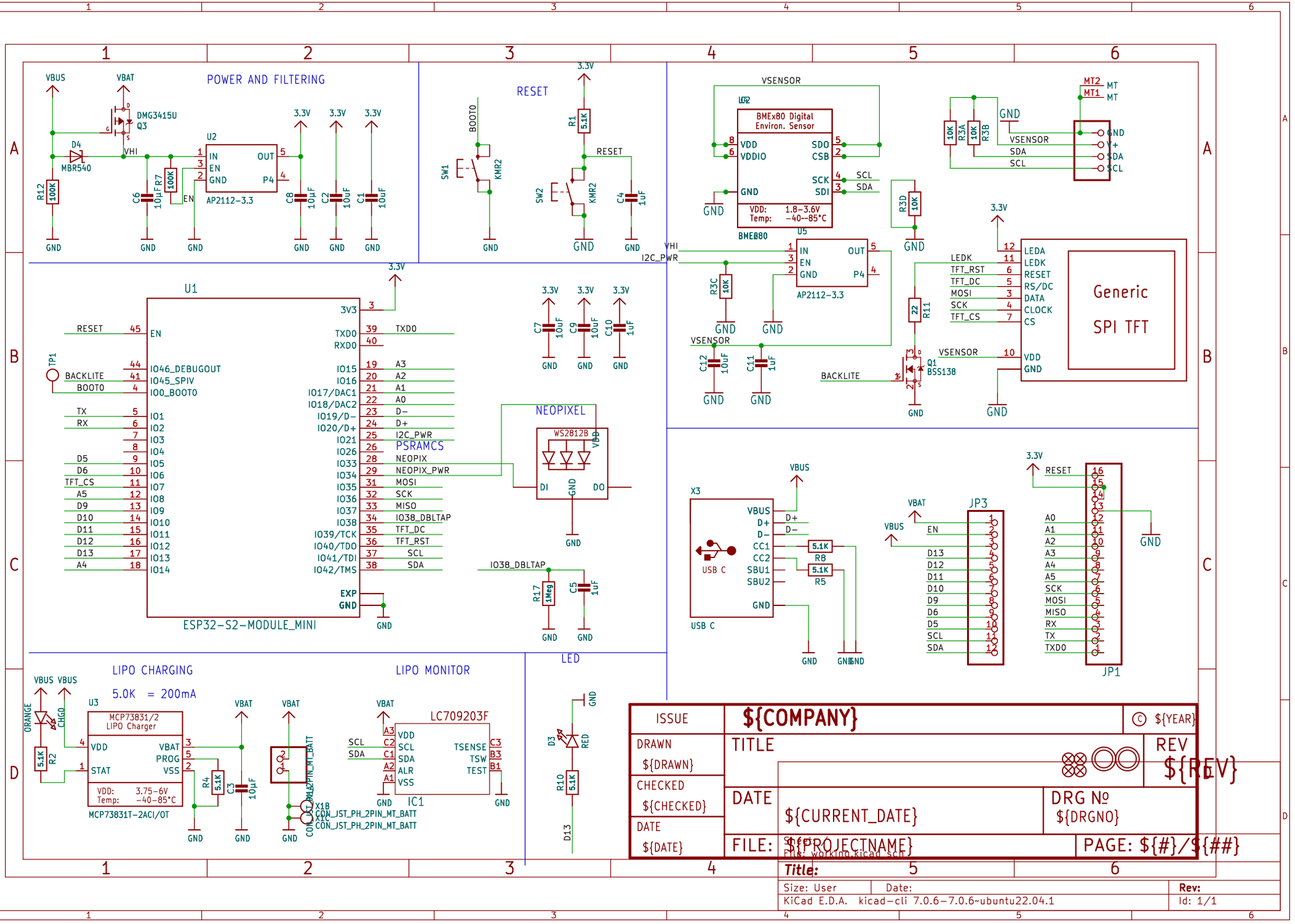
POWER AND FILTERING
VBUS
VBAT
DMG3415U
Q3
D4
MBR540
VHI
R12
100K
R7
100K
EN
U2
IN
EN
GND
OUT
P4
AP2112-3.3
C6
10µF
C8
10µF
C2
10uF
C1
10uF
3.3V
GND
RESET
BOOT0
SW1
KMR2
R1
5.1K
RESET
SW2
KMR2
C4
1uF
VSENSOR
U4
BME680
BMEx80 Digital
Environ. Sensor
VDD
VDDIO
GND
SDO
CSB
SCK
SDI
SCL
SDA
VDD: 1.8-3.6V
Temp: -40-85°C
R3A
10K
R3B
10K
R3D
10K
MT2
MT1
MT
GND
VSENSOR
SDA
SCL
U5
AP2112-3.3
VHI
I2C_PWR
R3C
10K
C12
10uF
C11
1uF
R11
22
Q1
BSS138
BACKLITE
LEDK
TFT_RST
TFT_DC
MOSI
SCK
TFT_CS
LEDA
RESET
RS/DC
DATA
CLOCK
CS
VDD
Generic
SPI TFT
U1
ESP32-S2-MODULE_MINI
EN
BACKLITE
BOOT0
TX
RX
D5
D6
TFT_CS
A5
D9
D10
D11
D12
D13
A4
IO46_DEBUGOUT
IO45_SPIV
IO0_BOOT0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
3V3
TXD0
RXD0
IO15
IO16
IO17/DAC1
IO18/DAC2
IO19/D-
IO20/D+
IO21
IO26
IO33
IO34
IO35
IO36
IO37
IO38
IO39/TCK
IO40/TDO
IO41/TDI
IO42/TMS
EXP
GND
A3
A2
A1
A0
D-
D+
I2C_PWR
NEOPIX
NEOPIX_PWR
MOSI
SCK
MISO
IO38_DBLTAP
TFT_DC
TFT_RST
SCL
SDA
PSRAMCS
C7
10uF
C9
10uF
C10
1uF
NEOPIXEL
WS2812B
VDD
DI
GND
DO
IO38_DBLTAP
R17
1Meg
C5
1uF
X3
USB C
VBUS
D+
D-
CC1
CC2
SBU1
SBU2
GND
5.1K
R8
5.1K
R5
JP3
JP1
RESET
A0
A1
A2
A3
A4
A5
SCK
MOSI
MISO
RX
TX
TXD0
LIPO CHARGING
5.0K = 200mA
U3
MCP73831/2
LIPO Charger
VDD
STAT
VBAT
PROG
VSS
VDD: 3.75-6V
Temp: -40-85°C
MCP73831T-2ACI/OT
ORANGE
CHGO
R2
5.1K
R4
5.1K
C3
10µF
LIPO MONITOR
IC1
LC709203F
A3
SCL
C2
SDA
C1
A2
ALR
A1
VSS
TSENSE
TSW
TEST
C3
B3
B1
X1B
X1C
CON_JST_PH_2PIN_MT_BATT
LED
D3
RED
R10
5.1K
D13
ISSUE
DRAWN
${DRAWN}
CHECKED
${CHECKED}
DATE
${DATE}
${COMPANY}
${YEAR}
TITLE
REV
${REV}
DATE
${CURRENT_DATE}
DRG №
${DRGNO}
FILE: ${PROJECTNAME}.kicad_sch
PAGE: ${#}/${##}
Title:
Size: User
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1
Id: 1/1